

LEVENT CELIKSAN

Senior AI & Software Engineer | Technical Founder of Sealify

Agentic AI · Computer Vision · AI Systems · Backend

Istanbul, Türkiye | Open to remote work | Open to relocating to the US (requires visa sponsorship)

leventceliksan@gmail.com | linkedin.com/in/levent-celiksan | github.com/LeventCeliksan | sealify.io

SUMMARY

Senior AI & Software Engineer and hands-on technical founder with **19+ years** of backend and systems engineering, including leading and mentoring developers. Took **Sealify** from idea to public launch as its sole technical founder. Focused on applied AI engineering: production systems validated by measurement on real-world web data rather than research prototypes — agentic AI, computer vision, multimedia matching, KYC/identity verification, and content authentication.

As AI makes code easier to write, the differentiator is deciding architecture, security, hardware capacity, unit cost, and sustainability — owned all of these at Sealify end to end, from build-vs-buy AI choices (open-source/local models vs. paid APIs, on-device vs. server-side inference) through production operations.

PROFESSIONAL EXPERIENCE

Technical Founder & Senior AI & Software Engineer — Sealify, Inc.

Nov 2023 – Present

Newark, Delaware (Remote) · Incorporated in Delaware in 2025

- **Launch:** Launched Sealify publicly in June 2026 as a multilingual (EN/TR/AR) SaaS platform with ~100 early users: Google sign-in, subscription plans (free to \$200/month) with crypto payments, a user dashboard, and an admin panel (users, credits, email, scheduled DMCA scans, user-approved takedown notices). Demo: youtu.be/LIoTl1nJU5c · Intro: youtu.be/Bh7kB-UIJhs
- **Measured results:** Validated web-wide content scanning, video matching, and autonomous social media archiving through **3,000+ measured test runs**: scanned **200,000+** candidate images and identified **10,000+ verified matches across 5,000+ domains** at **~98% precision** in manual review of scan results; a single video test found **1,050+ matches across 80+ domains**; archiving tests automatically processed and registered **10,000+ media items (450+ GB)**.
- **Reverse image/video search:** Built a multi-engine reverse search (Google Lens, Google Reverse Image, Yandex, Bing) that runs a free method in parallel with a paid API, failing over across API keys and methods when rate limits hit. From a single photo, poster, or video frame, it finds public copies across the web in a **median of 2.5 minutes**, including versions with changed resolution, removed or altered audio, or heavily altered colors or visual style.
- **Matching engine:** Engineered and calibrated a hybrid image and face matching engine (color/composition signals, InsightFace/ArcFace face gate, EfficientNet-B7 embeddings), raising matching accuracy from **~60% to ~97%** by tuning signal weights and thresholds; runs on parallel workers in a queue-based, multi-instance architecture.
- **Self-hosted KYC:** Built a two-stage identity verification pipeline — template-based recognition that accepts only passports, national IDs, and driver's licenses and rejects anything else, then a **blink-based liveness check** and open-source face matching of the selfie against the document photo — saving **~\$1 per verification** vs. paid providers such as Onfido or Jumio, with biometric data kept on its own infrastructure. Correctly verified **all 20 users** in a 20-person test; documents (including MRZ) can be uploaded or photographed, while the selfie is captured live by camera only (no uploads) to reduce spoofing and replay attacks.
- **Originality check & Bitcoin timestamping:** Before registration, videos pass an audio-first originality check (audio recognition, then frame search), and content already online is rejected. Approved content is anchored to Bitcoin via OpenTimestamps as a fingerprint manifest (SHA-256 + pHash of 27 frames × 12 crop/rotation variants), with verifiable certificates and no transaction fees for users. **1,600+ registrations** anchored on mainnet, each verifiable on public block explorers.
- **Autonomous monitoring:** Built worker systems that check Instagram, TikTok, and YouTube channels every 15 minutes, automatically downloading new uploads, extracting their fingerprints, and registering them on the blockchain; built a scraper that continuously monitors pirate movie sites and feeds newly listed titles into the detection pipeline in near real time.
- **AI research agent:** Built an agent with LangGraph and local LLMs that autonomously investigates, evaluates, and reports on suspected piracy sites end to end, generating evidence-based reports automatically.
- **Deepfake detection:** Developed a hybrid prototype that detects AI-generated images and videos: a decision engine combining image-analysis signals (regional blur anomalies, homogeneous texture, noise, color and contrast distribution) with voting across Hugging Face classifiers, plus an asynchronous video analysis service that samples frames from YouTube/TikTok videos.
- **NCII & deepfake discovery:** Built a victim-protection feature that finds non-consensual intimate and deepfake content by combining web image search (SerpApi/Yandex) with InsightFace/ArcFace face matching; shipped in paid plans, it scans adult content platforms with a privacy-focused flow that processes hundreds of images per scan with parallel downloads and Apple Silicon GPU acceleration, shows results blurred, and deletes data automatically.
- **Infrastructure, cost & business:** Optimized for long-term financial sustainability: planned RAM/CPU/GPU usage, server capacity, energy, and third-party service costs together; chose between AWS, GCP, Azure, Vultr, and self-hosted infrastructure by workload; handled Linux server security, backups, and monitoring. Bootstrapped, fully remote company on self-hosted hardware with no payroll, office, or paid LLM APIs: **~\$300/year** in fixed costs; led financial planning and investor talks.

Senior Software Engineer — BUBU (Kopuzlar Group) · Türkiye Python (Flask, FastAPI) APIs, e-commerce backends, servers, websites; led and mentored 2 developers; LLM/RAG prototypes (2023).	2018 – 2023
Software Engineer — Artworks Lab AB · Stockholm, Sweden (Remote) Boutique e-commerce agency: Django/DRF e-commerce backends scaled with Elasticsearch and Redis on AWS.	2015 – 2018
Software Engineer — Yellow Pages / DataWorks · United States (Remote) REST APIs, mapping features, and phone lookup service for a caller-ID mobile app (PHP/Laravel, MySQL, AWS).	2012 – 2015
Software Engineer — Jolly Tur · Istanbul, Türkiye Major Turkish travel company: PHP/MySQL booking and e-commerce backends, SOAP/XML supplier integrations, Linux/Windows servers.	2008 – 2011
Junior Software Engineer — REKLAMARKA · Türkiye Corporate websites, PHP/MySQL backends, admin panels, and e-commerce integrations.	2006 – 2008

PATENT APPLICATIONS

Inventor on **six US provisional patent applications** (USPTO, Nov–Dec 2025, Patent Pending) in AI-based content protection, deepfake detection, and multimedia matching, including: content tracking and protection (63/923,341); real-time deepfake detection (63/923,358); multimodal novelty assessment (63/923,342); autonomous video registration (63/934,369); multimedia semantic equivalence (63/946,066). Invited to **IPHatch North America 2026**, an open-innovation program for deep-tech startups.

SELECTED AI PROJECTS

Aesthetic Detector — BEAUTY (Android) <i>Personal product: Android app built in Kotlin with a credit-based revenue model; preparing for Google Play launch</i>	2025 – Present
- Fine-tuned a CLIP ViT-B/16-based model on 1,000+ portrait photos to classify signs of cosmetic intervention, then converted it to ONNX and optimized it to run inference entirely on-device, returning a result in ~2 seconds. - Developed a multi-stage scoring pipeline combining the overall classification score with EfficientNet-B0-based regional facial analysis; Google ML Kit for face detection and pose checks, ONNX Runtime for inference. - Privacy-oriented architecture: heavy AI processing runs on the user's device, so images never need to be sent to a server; the Node.js, Express, and MySQL backend handles only authentication, the credit system, data synchronization, and coordination.	

AI Advertising Agency Automation <i>Personal project: working demo</i>	2025 – Present
- Built a generative AI advertising agency with 7 agents (copywriter, photographer, designer, social media, developer, video, assistant) coordinated by a manager agent that turns natural-language chat requests into role-specific briefs on a shared SQLite task board, producing social posts, banners, campaign sites, and reels videos. - Built a single LLM gateway on Python, FastAPI, and Pydantic that picks Gemini model tiers by task importance, falls back automatically to local Ollama models on quota limits or errors, supports OpenRouter models, and tracks tokens and cost per job; designs are rendered from HTML/CSS with Playwright and videos encoded with ffmpeg locally, keeping server costs near zero. - Developed autonomous quality control: mechanical checks (missing files, banned words, font sizes measured before rendering), vision-model review of outputs at real mobile size for legibility and overflow, automatic prompt correction with up to 3 image regenerations, rotation across 12 layouts to prevent mode collapse, and revisions that rerun only the requested agent.	

TECHNICAL SKILLS

- Generative AI & agents:** LLMs, agentic AI, multi-agent systems, LangGraph, LangChain, tool/function calling, structured outputs, multi-LLM routing and failover, LLM evaluation, guardrails, Ollama, Gemini API, OpenRouter, model fine-tuning
- RAG & retrieval:** RAG, embeddings, semantic search, vector search (FAISS)
- ML & computer vision:** PyTorch, Hugging Face Transformers, ONNX / ONNX Runtime, on-device inference, OpenCV, InsightFace/ArcFace, face matching, liveness detection, EfficientNet, CLIP, DINOv2, YOLOv8, NumPy, Pandas, scikit-learn, perceptual hashing, deepfake detection, document forensics (MRZ), OCR/Tesseract, Whisper
- Backend & data:** Python (FastAPI, AsyncIO, Pydantic, Flask, Django/DRF, SQLAlchemy, Celery, pytest), Node.js/Express, REST, WebSockets, JWT; MySQL, PostgreSQL, MongoDB, SQLite, Redis, Elasticsearch, RabbitMQ
- Mobile, automation & blockchain:** Android (Kotlin, Google Play Billing, ML Kit); Playwright, Chrome DevTools Protocol, yt-dlp, ffmpeg, SerpApi; OpenTimestamps / Bitcoin anchoring, SHA-256
- Cloud & DevOps:** AWS, GCP, Azure, Vultr, self-hosted; Linux, Nginx, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab CI, Jenkins), Prometheus, Grafana, Sentry, Cloudflare; capacity planning and cost modeling
- Working knowledge:** Model Context Protocol (MCP), pgvector/Qdrant, hybrid retrieval, reranking, AI observability and tracing, microservices, event-driven architecture, distributed systems, OAuth2/RBAC

EDUCATION & LANGUAGES

Collège LaSalle, Montréal, Canada — Bachelor's Degree, Computer Science Technology	2002 – 2006
Languages: Turkish — Native English — Professional working proficiency	